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(54) **VEHICLE NETWORK AND METHOD OF COMMUNICATION**

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H04J 3/06 (2006.01)

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See application file for complete search history.

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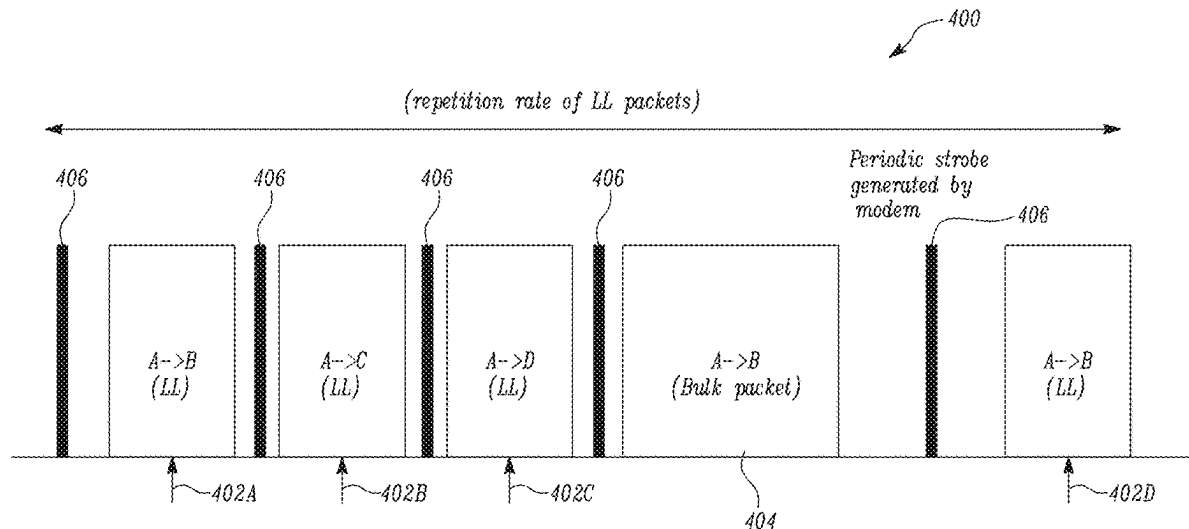
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(57) **ABSTRACT**

A method of communication in a vehicle network is provided. An example method includes transmitting a network allocation map in a TDMA cycle, indicating reservation of time slots in the TDMA cycle. The method further includes transmitting a synchronization signal in the TDMA cycle, to synchronize the timing of nodes in the vehicle network. Each of the reserved time slots is identified by at least a network ID of a transmitting node in the vehicle network, and a slot type comprising one of a low latency traffic slot, and a bulk traffic slot. Further, the low latency traffic slots are repeated in the TDMA cycle at least as frequently as a guaranteed QoS latency parameter. Further, the bulk traffic slots are at least as long as a guaranteed QoS throughput parameter.

18 Claims, 4 Drawing Sheets



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division of application No. 16/659,374, filed on Oct. 21, 2019, now Pat. No. 11,539,638.

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(52) **U.S. Cl.**

CPC **H04L 12/40163** (2013.01); **H04L 47/722** (2013.01); **H04L 67/12** (2013.01)

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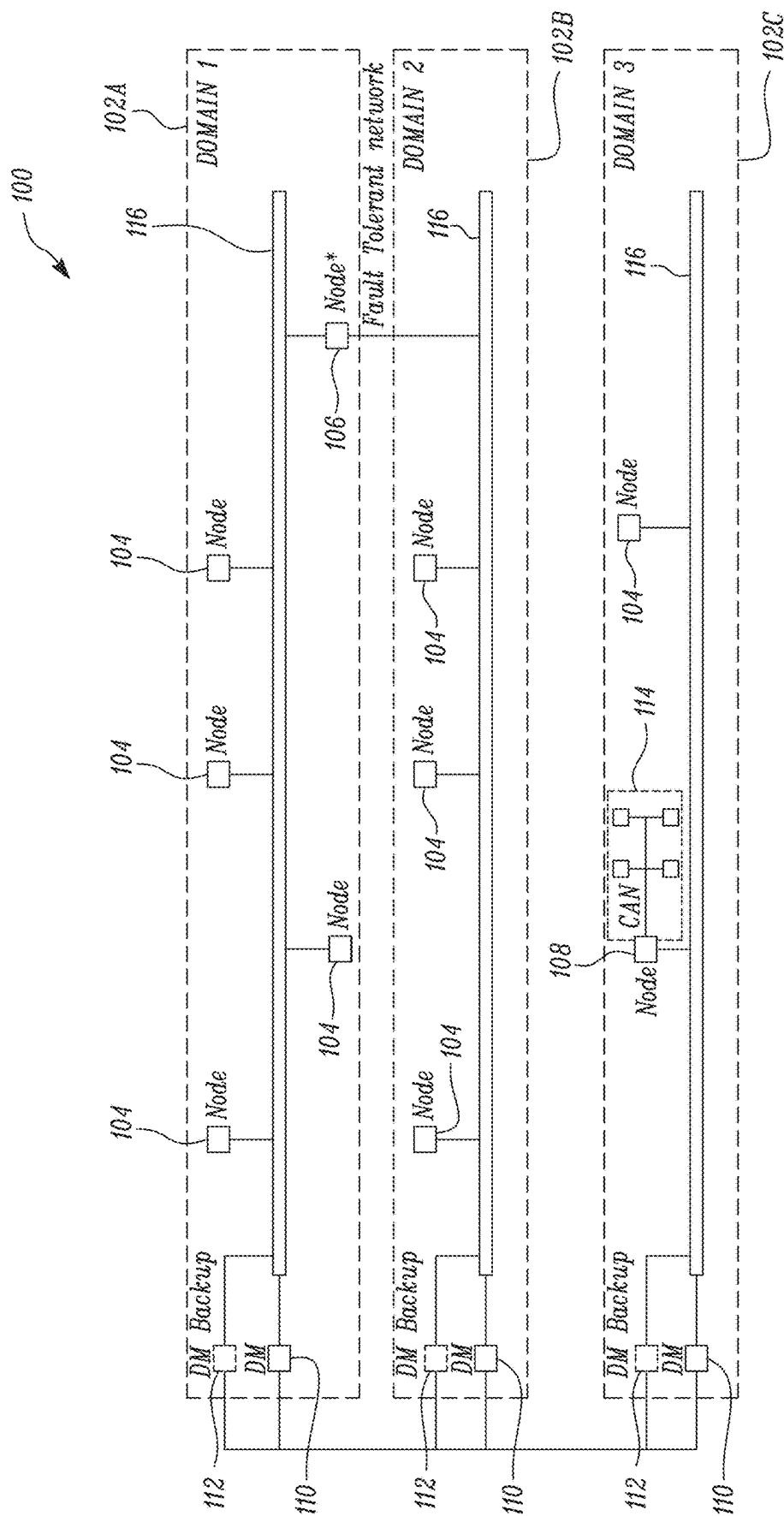


FIG. 1

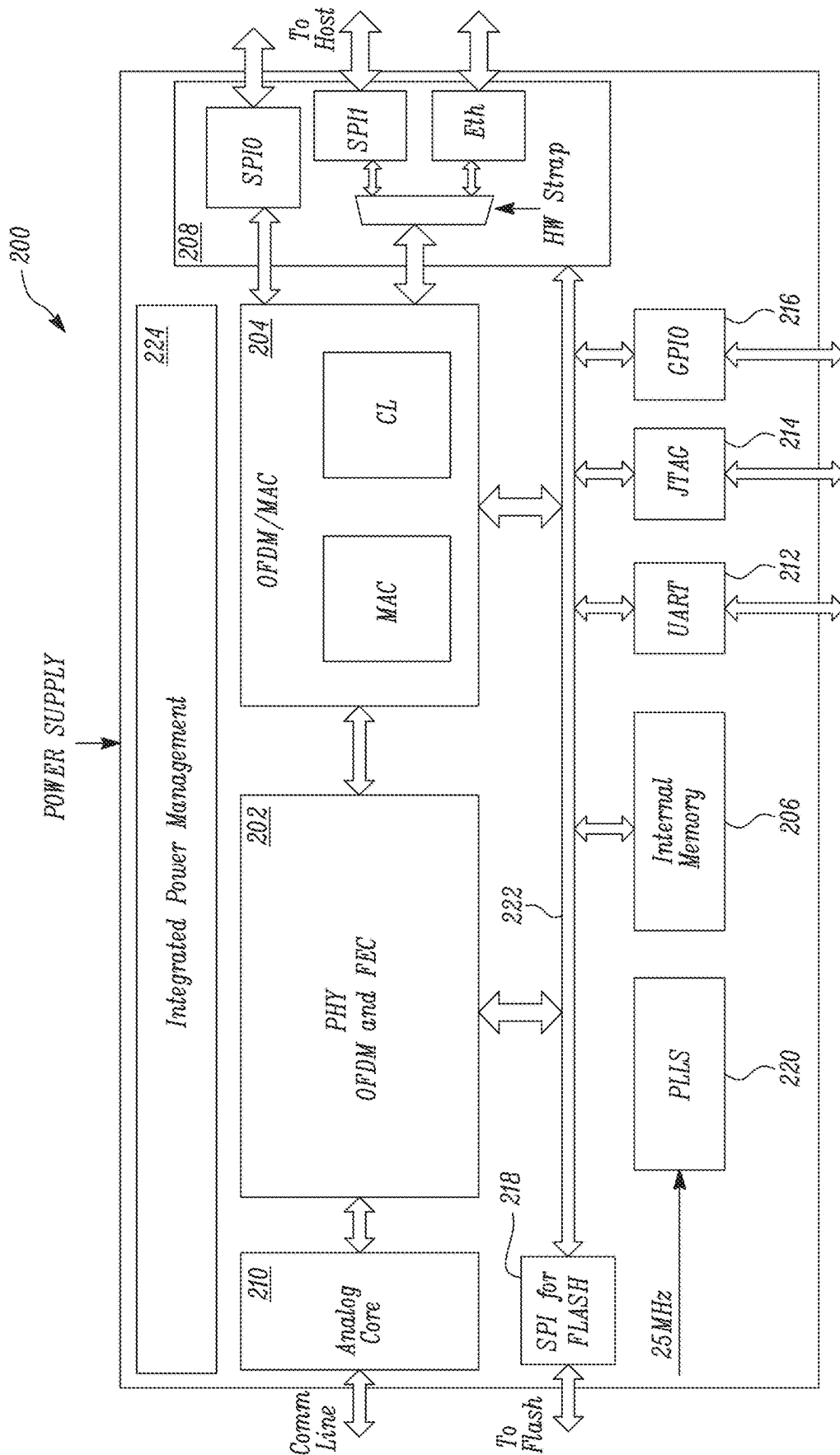


FIG. 2

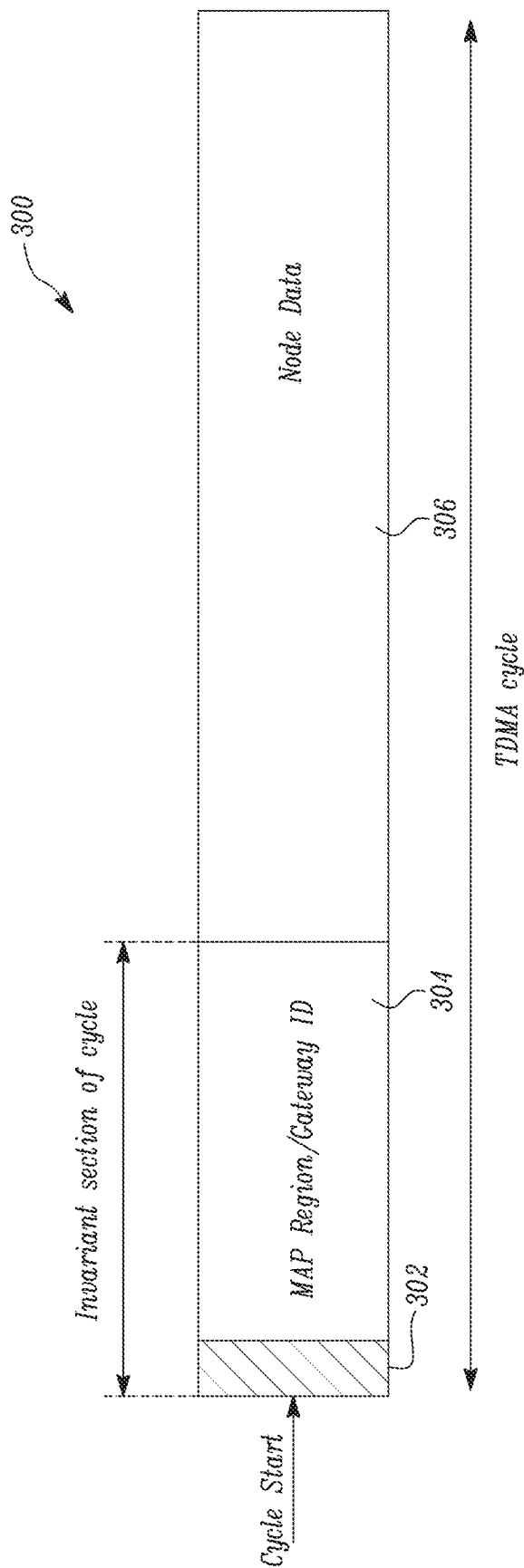


FIG. 3

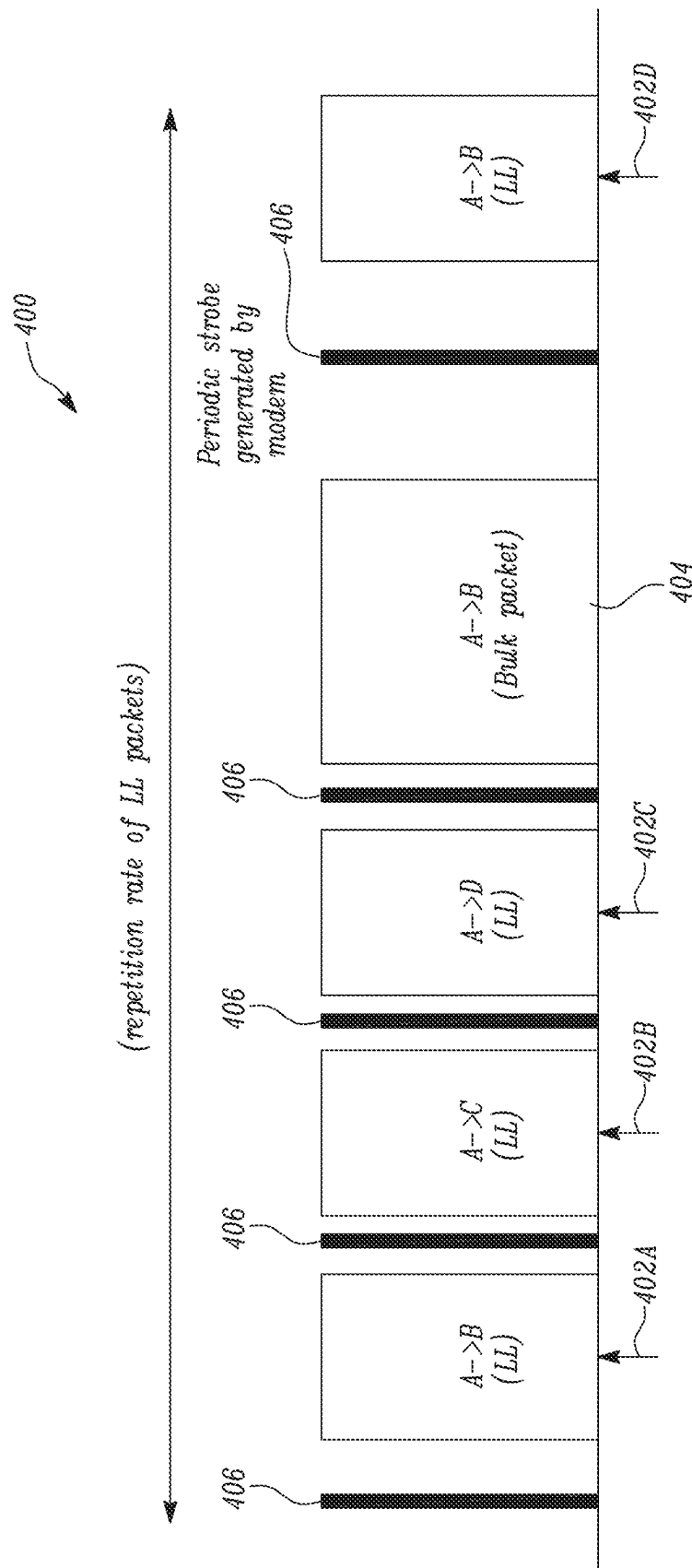


FIG. 4

1

VEHICLE NETWORK AND METHOD OF COMMUNICATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/145,543, filed on Dec. 22, 2022, which is a divisional of U.S. patent application Ser. No. 16/659,374, filed on Oct. 21, 2019, titled "VEHICLE NETWORK AND METHOD OF COMMUNICATION," which claims priority to U.S. Provisional Patent Application No. 62/748,894, filed Oct. 22, 2018, titled "VEHICLE NETWORK AND METHOD OF COMMUNICATION," all of which are hereby incorporated by reference in their entirety and for all purposes.

BACKGROUND

Technical Field

The present disclosure relates to a vehicle network and method of communication employed therein.

Description of Related Art

Traditional vehicular networks have a large number of point-to-point links between the vehicle computer and various vehicular subsystems. Typically, wiring harnesses for such a topology are inherently complex, and often difficult to install and troubleshoot. Modern vehicles have different categories of data traffic, such as sensor data, actuator and control signaling, as well as multimedia traffic. Sensor and control signaling typically requires low latency and has a low volume of data. Accordingly, sensor and control signaling data needs to be communicated more frequently, in smaller communication units (e.g. data packets or time slots). In contrast, data gathered from the sensors themselves, such as multimedia traffic, requires larger communication units (e.g. data packets or time slots) but may not be critical to the operation of the vehicle. Thus, multimedia traffic may be seen as low priority, bulk traffic. It is recommended that the different classes of communication traffic be differentiated from one another to ensure that each data traffic type meet quality of service (QOS) criteria, for proper operation of the vehicle systems.

Hence, there is a need for a vehicle communication network and communication method to alleviate these and other drawbacks of the state of the art.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a vehicle communication network according to certain embodiments.

FIG. 2 illustrates a functional block diagram of a modem used in the vehicle communication network, according to certain embodiments.

FIG. 3 illustrates a logical TDMA cycle structure, according to certain embodiments.

FIG. 4 illustrates a physical TDMA cycle structure, according to certain embodiments.

DETAILED DESCRIPTION

The present disclosure relates to a vehicle onboard network system, and methods of communication deployed between the various components of the network. In some

2

embodiments, the vehicle onboard network may enable transmission, and routing, of information from sensors, devices, modules, systems, processing elements, and similar components, located within a vehicle (herein referred to as 'nodes'). For example, a sensor may include a temperature sensor located within the vehicle. A temperature measured by this sensor may be transmitted via the vehicle onboard network to another location (e.g., a central processing system or a temperature control system). As another example, a sensor may be associated with a door of the vehicle. In this example, the sensor may indicate a position of the door (e.g., whether the door is closed or an angle of the door). The vehicle onboard network described herein may enable such transmission using decreased complexity as compared to prior techniques. Additionally, the vehicle onboard network may allow for higher transmission rates of other types of data as compared to prior techniques.

An example prior technique used to transmit information in a vehicle includes use of a controller area network (CAN) bus. The CAN bus may enable microcontrollers and devices to communicate, for example based on a message-based protocol, however it introduces technical deficiencies. For example, a CAN bus only operates at data transmission rates that may not be acceptable for modern vehicles needing to communicate at high speed to control various components. In this example, data transmission rates on a CAN bus may be limited to 1 megabit per second. Additionally, there are substantial wiring and processing complexities associated with use of the CAN bus.

A vehicle onboard network described herein allows for higher data transmission rates than a CAN bus while advantageously reducing wiring complexity. In some embodiments, an improved bus may be used which is shareable between a multitude of nodes. For example, the improved bus may be shareable by a subset, or all, of the nodes included in a vehicle. In this example, the improved bus may be shareable between nodes included in a same group of nodes (e.g., a domain as described below). To ensure that the improved bus is being efficiently used, the vehicle onboard network may use time division multiple access (TDMA) techniques. In some embodiments, a network allocation map may be used to inform times at which a node can transmit information via the improved bus. As will be described, the network allocation map may comprise TDMA slot reservations assignable to nodes. Thus, the vehicle onboard network may allow for the timely, and reliable, transmission of information from multitudes of nodes.

Advantageously, additional nodes may be rapidly added to the improved bus to enhance a vehicle over time. For example, a sensor may be added to a vehicle which is usable to inform safety associated with seatbelts. In this example, the sensor may be communicatively coupled to the improved bus and its information transmitted using the TDMA network described herein. For example, the sensor may transmit information according to a network allocation map which is shared amongst other nodes communicatively coupled to the improved bus.

In this way, the complexity associated with sharing a bus of one or more electrical connections (e.g., wires) may be reduced through the above-described TDMA slot reservations. For example, since use of the improved bus may be controlled, at least in part, using a network allocation map, an improved bus may more simply be formed using a threshold number of wires (e.g., 4, 6, and so on). These wires may then be routed throughout a vehicle or throughout a

particular domain of the vehicle. Thus, wiring and processing complexity may be reduced as compared to more complex prior techniques.

As described above, the vehicle onboard network according to various embodiments of the present invention may be a time division multiple access (TDMA) network. Data over the TDMA vehicle onboard network is transmitted in data packets having time slots of predetermined, fixed durations. Nodes in the vehicle onboard network are synchronized to a central clock. Such a synchronous, reserved TDMA communication scheme allows guaranteed QoS communications, avoiding packet collisions, and competing transmissions in the same time slots.

FIG. 1 illustrates a vehicle communication network 100 according to certain embodiments. The vehicle communication network 100 is organized into a number of domains 102A, 102B, and 102C. Each domain 102 may represent a logical grouping of several communication nodes which provides an improved bus for transferring data within a vehicle. For example, one scheme of logical grouping in a vehicle may be grouping nodes according to vehicle component location such as, but not limited to, vehicle front left, vehicle front right, vehicle cabin left side, vehicle cabin right side, vehicle trunk, and so forth. Another example logical grouping may be according to vehicle subsystem (e.g., air conditioning, temperature, certain safety systems, exterior lighting, interior lighting, and so on). Other schemes of logical groupings to form domains in the vehicle communication network 100 are also envisioned and fall within the scope of the present disclosure. Multiple domains 102 may be connected to each other using bridging methods for data networks.

Each domain 102 includes one or more nodes 104. Each node 104 may be a vehicle module which communicates with other modules in the vehicle. Examples of vehicle modules include, doors, seats, battery pack, front end module, front subframe module, rear subframe module, and so forth. Each module may include various sensors, and/or actuators—for example, impact sensors, pressure sensors, temperature sensors, wheel speed sensors, airbag inflators, HVAC systems, traction power electronics, and so forth.

Some nodes may be critical to the safe operation of the vehicle, such as airbag systems, traction power electronics, vehicle stability systems, and the like. Such nodes, such as node 106 may be connected in a fault tolerant configuration, having a physical connection with two separate domains. This configuration provides failover capability in the event that the node 106 may not be able to communicate with the other vehicle systems, over one domain.

Some nodes 108 may also include layered subnets, comprising further child nodes, which may communicate with node 108 transparent to the vehicle communication network 100. Such a layered subnet may be beneficial, for example, to group modules in a particular vehicle location, such as various sensors and electronics in the door. Typically, such a layered subnet may be deployed, for example, in vehicle modules that do not communicate often, or only communicate data less than the minimum data time slot available in the vehicle communication network 100.

Each domain may have one domain master 110, and a backup domain master 112. The domain master 110 is a special type of node for managing the communication resources in the vehicle communication network 100 and may implement medium access control (MAC) protocol. The backup domain master 112 has identical, or near iden-

tical, capabilities compared to the domain master 110, and is designated as a failover node, in the event that a domain master 110 fails.

The nodes 104, 106, 108, domain master 110, and backup domain master 112, are all interconnected over a single physical conductive medium 116 within the domain. In various embodiments, the single physical conductive medium 116 may be vehicle DC power delivery conductors. For example, the vehicle communication network 100 may be implemented as a power line communication (PLC) network. In this example, the network 100 may be provided via power connections within the vehicle (e.g., DC power connections).

Each node 104, 106, and 108 includes various vehicle sensors and actuators, and a modem for communicating with the other nodes over the vehicle communication network 100.

FIG. 2 illustrates a functional block diagram of a modem 200 used in the vehicle communication network 100, according to certain embodiments. The modem 200 supports two categories of traffic: a low latency traffic, which typically has low throughput (for example, not exceeding 1 Mbps), and a low latency guarantee (for example, 500 microseconds or less); and a high throughput traffic (for example, 50 Mbps), with a lower latency requirement (for example, 40 milliseconds). In other embodiments, the modem 200 supports three or more categories of traffic.

The modem 200 includes a physical layer processor 202, a MAC processor 204, an internal memory 206, and various interfaces. The interfaces include a host interface 208, a comm. analog core 210, Universal Asynchronous Receiver Transmitter (UART) 212, a Joint Test Action Group (JTAG) interface 214, a general purpose I/O (GPIO) interface 216, and a flash interface 218. The modem 200 also includes a phase locked loop (PLL) module 220, for synchronizing the modem timing with a central clock. The various processors and interfaces of the modem communicate with each other over a common bus fabric 212. The modem includes an integrated power management module 214, which provides the power to the various processors and interfaces of the modem.

The physical layer processor 202 may be an OFDM processor according to various embodiments. The physical layer processor 202 may include, for example, an Inverse Fast Fourier Transform (IFFT) module and a forward error correction (FEC) module. In various embodiments, the OFDM processor operates in a passband channel of 2 MHz to 50 MHz. However, other channels are also envisioned depending on the designed network throughput and channel loss.

In one embodiment, the physical layer processor 202 supports low latency packets, for which the payload is encoded in the packet header. Such low latency packets are typically small in size. An exemplary low latency packet includes 16 bytes or 32 bytes of payload data, and 3 bytes for source ID, destination ID, and packet type field. In the exemplary channel of 2 MHz to 50 MHz, the example 35-byte low latency packet may require a time slot smaller than or equaling 25 microseconds.

The MAC processor 204 handles the MAC layer processing of the communications. The MAC processor 204 may include a MAC layer processing unit and a cross layer processing unit. The MAC processor 204 may implement a TDMA based MAC layer. In various embodiments, the MAC processor 204 is configured to operate in a lean mode, wherein multiple physical layer frames from different nodes 104 can be assembled with minimal MAC overhead. To

5

further improve latency performance, the MAC processor **204** may support transmission of data frames with and without an acknowledgement mechanism (e.g. ACK responses). In embodiments where ACK responses are required for transmissions, the ACK frame may be designed to be short, such as 2 OFDM symbols, for example.

In various embodiments, the MAC processor **204** may have specialized hardware to support low latency communication packets, thus further improving the latency parameters for the low latency traffic.

The internal memory **206** may include an SRAM memory or a DRAM memory. The internal memory **206** is the primary memory for various operations and data buffering for the physical layer processor **202**, and the MAC processor **204**. The internal memory **206** may also include a flash memory for saving persistent data, such as the TDMA network allocation maps, for example. In embodiments, the internal memory **206** is a flash memory. The internal memory **206** may store a network ID of the node of which modem **200** is a part. The network ID of the node of which modem **200** is a part, may be transmitted by the host controller (or hub), at the time of initial setup, or reconfiguration.

The host interface **208** may include a Serial Peripheral Interface (SPI) for communicating with a central host controller (not illustrated; also referred to as the hub, herein). In embodiments, the host interface **208** may be used by the host to signal TDMA network allocations maps to the modem **200**, and for pushing firmware updates to the modem **200**. The host interface **208** may include dual SPI-SPI0 and SPI1. In some embodiments, the dual interfaces may be deployed as redundant interfaces for failover modes. In other embodiments, the dual interfaces may be deployed as supplementary interfaces, such that SPI0 is the primary interface, and SPI1 may be used for additional traffic, when required.

The host interface **208** may also include an ethernet interface, implemented using a reduced gigabit media independent interface (RGMII). The RGMII ethernet interface may thus allow coupling of the MAC processor **204** with an external physical layer ethernet chip (not illustrated), independent of the medium (e.g. twisted pair, or coaxial, or fiber optic, and the like).

The comm. analog core **210** is a front-end module for coupling with the external network cables or conductors (illustrated as comm. line in FIG. 2). In some embodiments, the comm. analog core **210** conforms to the Power Line Communication (PLC) communication standard. The comm. analog core **210** may support differential signaling, or single ended signaling, or both with the ability to configure the signaling mode. For example, the modem **200** may be able to switch between a differential signaling mode and a single ended signaling mode based on whether the modem is transmitting/receiving a low latency data packet, or a high throughput bulk data packet, or based on the line conditions of the network cable/conductor.

In some embodiments, the comm. analog core **210** may couple to an external line driver. In other embodiments, the comm. analog core **210** may include the line drivers, such that no external hardware is required to amplify the signals further. Further, termination resistors may either be integrated into the comm. analog core **210** or may be connected externally to the comm. analog core **210**.

The modem **200** may also leverage other interfaces, such as UART **212**, JTAG **214**, and GPIO **216** for testing and debugging purposes.

Flash interface **218** may be used for flashing persistent data for the modem **200**. Such persistent data may include

6

TDMA network allocation maps, configuration information for analog core **210**, and so forth. The flash interface **218** may be an SPI in accordance with one embodiment. The flash interface **218** may be communicably coupled to the host controller (or hub), or to an external port for connecting a flash programming tool, or both.

The modem **200** also includes one or more PLLs **220**. The PLL **220** is configured to maintain synchronization with the host controller (or hub), and consequently with all other nodes **104** of the vehicle communication network **100**. In some embodiments, the PLL **220** may receive a fixed frequency clock signal, such as a 25 MHz clock signal, from the host controller (or hub) to maintain the timing synchronization by adjusting a timing circuitry of the node, based on the received synchronization signal.

The various processors, interfaces, and memory for the modem **200** may be communicably coupled to one another via a bus fabric **222**. In some embodiments, the modem **200** may communicate via the bus fabric **222** at particular times. For example, the modem **200** may use the network allocation map to provide information via the bus fabric **222**.

The integrated power management module **224** is a power controller for providing power to the various processors, interfaces, and memories of the modem **200**. The integrated power management module **224** may also provide control signals to switch the modem **200** to a power saving mode, or a sleep mode when required.

In some embodiments, a modem **200** may receive a network allocation map in a TDMA cycle, and identify from the network allocation map, particular time slots reserved for transmission by the associated node. The modem **200** may access the internal memory **206**, and compares a received network ID to the one stored in memory **206** to identify whether the TDMA time slot has been assigned to itself.

The modem **200** buffers the data to be transmitted beforehand, and at the identified time slot reservation, the modem **200** transmits the buffered data.

If a modem **200** does not identify a time slot reservation for transmission by its node, the modem excludes itself from transmitting the buffered data. This implementation may be helpful for isolating faults, and bringing faulty nodes offline, so as not to interfere with the operation of the rest of the nodes in the vehicle.

FIG. 3 is a logical representation of an exemplary TDMA cycle **300** to be employed in various embodiments. The TDMA cycle may include a cycle start indicator **302**, a MAP region **304**, and a portion for node data **306**. The cycle start indicator **302**, and the MAP region **304** together form the invariant section of the cycle and repeat exactly for every TDMA cycle **300** transmitted in the vehicle communication network **100**.

The cycle start indicator **302** is a fixed symbol indicating the beginning of the TDMA cycle **300**. The cycle start indicator may be known to all nodes **104** such that the nodes **104** can easily detect the start of a TDMA cycle **300**. The cycle start indicator **302** may also function as a failsafe for ensuring synchronization of the nodes **104** (e.g. modems **200**) with the host controller (or hub). If a node **104** (or modem **200**) does not detect the known TDMA cycle start indicator **302** at its designated time slot, the modem **200** may ascertain that it is out of synchronization with the host controller (or hub), and may initiate a synchronization procedure to regain synchronization with the host controller (or hub).

The MAP region **304** allocates TDMA time slots for the entire TDMA cycle **300**. The MAP region **304** includes a TDMA network allocation map, and a gateway ID supplied

by the host controller (or hub). The TDMA network allocation map reserves the TDMA time slots in the region for node data **306**, for nodes **104** to communicate with one another over the vehicle communication network **100**. The TDMA network allocation map defines which pair of nodes communicates in a given TDMA time slot, the type of traffic of the TDMA time slot (whether for low latency traffic, or for high throughput bulk traffic), and the duration for the TDMA time slot allocation.

According to various embodiments of the present invention, a host controller or hub reserves TDMA time slots for various nodes **104** in the vehicle communication network **100**. The host controller or hub then transmits the TDMA time slot allocation to all the nodes **104** in the network, using a network allocation map.

A network allocation map comprises TDMA slot reservations, wherein each of the reserved time slots is identified by a network ID of a transmitting node **104**, a network ID of a receiving node, a slot type indicating whether the TDMA time slot allocation is a low latency traffic slot, or a bulk traffic slot, and a duration of the TDMA time slot allocation. The duration of the TDMA time slot allocation indicates the number of contiguous time allocation periods to be used for sending data from a source Node to a destination Node (e.g., via a bus, such as a common bus).

The low latency traffic slot slots are repeated in the TDMA cycle at least as frequently as a guaranteed QoS latency parameter. The high throughput bulk traffic slots are at least as long as a guaranteed QoS throughput parameter.

Network allocation maps may be in a binary format, in accordance with various embodiments. In one embodiment, the maximum size of network allocation maps may be 4,800 bytes or fewer. The network allocation maps may also include a map ID, which indicates which particular one of several network allocation maps is being used. To switch maps, the host controller may simply transmit a map ID instead of the complete network allocation map.

To reduce network allocation map size, the vendor may choose to partition the map so that low latency slots, which repeat every 500 μ s, are specified only once in the map. Network allocation maps may also include a mechanism for modem **200** that receives it to validate the integrity of the received network allocation map.

The map region **304** is also invariant within the TDMA cycle **300**. This allows nodes **104** to join the vehicle communication network **100** at any given point, since the nodes **104** know a priori where the TDMA network allocation map is located with respect to the beginning of the TDMA cycle **300**.

The map region **304**, although illustrated as a contiguous group of TDMA time slots in the logical representation of the TDMA cycle **300**, in a physical implementation, the map region **304** may occupy TDMA time slots distributed throughout a physical TDMA cycle.

The region for node data **306** comprises TDMA slots for data communication by the nodes **104**. The node data **306** includes low latency slots, as well as high throughput bulk data slots.

An example arrangement of the node data **306** is described in conjunction with FIG. 4 below.

FIG. 4 represents a part **400** of a physical TDMA cycle. The part **400** may be referred to as a mini-cycle **400**. The physical TDMA cycle may be constituted of a large number of such mini-cycles **400**. In one example embodiment, a mini-cycle **400** may be of 500 microseconds duration, and a physical TDMA cycle may be made of 80 such mini-cycles **400**, for a total of a 40 millisecond TDMA cycle.

The mini-cycle **400** includes low latency slots **402A**, **402B**, **402C**, and **402D**, high throughput bulk data slots **404**, and a number of strobe signals **406**.

The low latency slots are time slots having short duration (for example, 25 microseconds or less), high repetition (for example, repeated once every 500 microseconds), since the low latency packets are required to exhibit a guaranteed maximum latency. In other words, in this example implementation, the low latency slots guarantee a maximum latency of 500 microseconds before the same pair of nodes may communicate a second time.

The high throughput bulk data slots are time slots having a longer duration (for example, 37.5 microseconds), carry a larger amount of data per time slot (for example, 112 bytes of node data). However, the high throughput bulk data slots are only repeated less frequently (for example, once every 40 milliseconds), since a higher latency is tolerable for the high throughput traffic.

The strobe signals **406** may be used by the modem **200** for performing automatic gain control.

What is claimed is:

1. A system, comprising:

one or more processors configured to:

determine a network allocation map in a time division multiple access (TDMA) cycle, the network allocation map comprising a plurality of TDMA slot reservations that are assignable corresponding nodes from among a plurality of nodes in a vehicle communication network;

determine assignments for each of the nodes of the plurality of nodes in the vehicle communication network, the assignments comprising assignment to low latency traffic slot reservations or bulk traffic slot reservations, wherein the low latency traffic slot reservations have a first latency guarantee of 500 microseconds or less, and the bulk traffic slot reservations have a second latency guarantee of 40 milliseconds or less;

provide the network allocation map to a first node and a second node from among the plurality of nodes, the first node assigned to a low latency traffic slot reservation having a first latency guarantee that satisfies a first threshold, and

the second node assigned to a bulk traffic slot reservation having a second latency guarantee that satisfies a second threshold; and

cause one or more messages to be transmitted by the plurality of nodes in accordance with the low latency traffic slot reservations and the bulk traffic slot reservation.

2. The system of claim 1, wherein the one or more processors configured to determine the assignments for each of the nodes of the plurality of nodes are configured to:

determine the assignments for each node of the plurality of nodes based on a type of data associated with the one or more messages to be transmitted by each node of the plurality of nodes.

3. The system of claim 1, wherein the first node is associated with a first sensor configured to generate first sensor data during operation of the vehicle, the first sensor data associated with a first type that is transmitted during the low latency traffic slot reservation, and

wherein the second node is associated with a second sensor configured to generate second sensor data during operation of the vehicle, the second sensor data associated with a second type that is transmitted during the bulk traffic slot reservation.

4. The system of claim 1, wherein the first node is configured to transmit messages having a first size that satisfy a first size threshold associated with the low latency traffic slot reservation; and

wherein the second node is configured to transmit messages having a second size that is greater than the first size, the second size satisfying a second size threshold associated with the bulk traffic slot reservation.

5. The system of claim 1, wherein the one or more processors configured to provide the network allocation map to the first node and the second node are configured to:

provide the network allocation map to the first node in a first domain, and

provide the network allocation map to the second node in a second domain.

6. The system of claim 5, wherein the one or more processors are further configured to:

provide the network allocation map to a third node, the third node being in a third domain, wherein the first node and the third node are connected and establish a fault tolerant configuration.

7. A non-transitory computer-readable medium storing instructions thereon that, when executed by one or more processors, cause the one or more processors to:

determine a network allocation map in a time division multiple access (TDMA) cycle, the network allocation map comprising a plurality of TDMA slot reservations that are assignable corresponding nodes from among a plurality of nodes in a vehicle communication network;

determine assignments for each of the nodes of the plurality of nodes in the vehicle communication network, the assignments comprising assignment to low latency traffic slot reservations or bulk traffic slot reservations, wherein the low latency traffic slot reservations have a first latency guarantee of 500 microseconds or less, and the bulk traffic slot reservations have a second latency guarantee of 40 milliseconds or less;

provide the network allocation map to a first node and a second node from among the plurality of nodes, the first node assigned to a low latency traffic slot reservation having a first latency guarantee that satisfies a first threshold, and

the second node assigned to a bulk traffic slot reservation having a second latency guarantee that satisfies a second threshold; and

cause one or more messages to be transmitted by the plurality of nodes based on the assignment of each node to the low latency traffic slot reservations and the bulk traffic slot reservations.

8. The non-transitory computer-readable medium of claim 7, wherein the instructions that cause the one or more processors to determine the assignments for each of the nodes of the plurality of nodes cause the one or more processors to:

determine the assignments for each node of the plurality of nodes based on a type of data associated with the one or more messages to be transmitted by each node of the plurality of nodes.

9. The non-transitory computer-readable medium of claim 7, wherein the first node is associated with a first sensor configured to generate first sensor data during operation of the vehicle, the first sensor data associated with a first type that is transmitted during the low latency traffic slot reservation, and

wherein the second node is associated with a second sensor configured to generate second sensor data during operation of the vehicle, the second sensor data asso-

ciated with a second type that is transmitted during the bulk traffic slot reservation.

10. The non-transitory computer-readable medium of claim 7, wherein the first node is configured to transmit messages having a first size that satisfy a first size threshold associated with the low latency traffic slot reservation; and wherein the second node is configured to transmit messages having a second size that is greater than the first size, the second size satisfying a second size threshold associated with the bulk traffic slot reservation.

11. The non-transitory computer-readable medium of claim 7, wherein the instructions that cause the one or more processors to provide the network allocation map to the first node and the second node cause the one or more processors to:

provide the network allocation map to the first node in a first domain, and

provide the network allocation map to the second node in a second domain.

12. The non-transitory computer-readable medium of claim 11, wherein the instructions further cause the one or more processors to:

provide the network allocation map to a third node, the third node being in a third domain, wherein the first node and the third node are connected and establish a fault tolerant configuration.

13. A method, comprising:

determining, by one or more processors, a network allocation map in a time division multiple access (TDMA) cycle, the network allocation map comprising a plurality of TDMA slot reservations that are assignable corresponding nodes from among a plurality of nodes in a vehicle communication network;

determining, by the one or more processors, assignments for each of the nodes of the plurality of nodes in the vehicle communication network, the assignments comprising assignment to low latency traffic slot reservations or bulk traffic slot reservations, wherein the low latency traffic slot reservations have a first latency guarantee of 500 microseconds or less, and the bulk traffic slot reservations have a second latency guarantee of 40 milliseconds or less;

providing, by the one or more processors, the network allocation map to a first node and a second node from among the plurality of nodes,

the first node assigned to a low latency traffic slot reservation having a first latency guarantee that satisfies a first threshold, and

the second node assigned to a bulk traffic slot reservation having a second latency guarantee that satisfies a second threshold; and

causing, by the one or more processors, one or more messages to be transmitted by the plurality of nodes based on the assignment of each node to the low latency traffic slot reservations and the bulk traffic slot reservations.

14. The method of claim 13, wherein determining the assignments for each of the nodes of the plurality of nodes comprises:

determining, by the one or more processors, the assignments for each node of the plurality of nodes based on a type of data associated with the one or more messages to be transmitted by each node of the plurality of nodes.

15. The method of claim 13, wherein the first node is associated with a first sensor configured to generate first sensor data during operation of the vehicle, the first sensor

data associated with a first type that is transmitted during the low latency traffic slot reservation, and

wherein the second node is associated with a second sensor configured to generate second sensor data during operation of the vehicle, the second sensor data associated with a second type that is transmitted during the bulk traffic slot reservation. 5

16. The method of claim **13**, wherein the first node is configured to transmit messages having a first size that satisfy a first size threshold associated with the low latency traffic slot reservation; and 10

wherein the second node is configured to transmit messages having a second size that is greater than the first size, the second size satisfying a second size threshold associated with the bulk traffic slot reservation. 15

17. The method of claim **13**, wherein providing the network allocation map to the first node and the second node comprises:

providing, by the one or more processors, the network allocation map to the first node in a first domain, and 20 providing, by the one or more processors, the network allocation map to the second node in a second domain.

18. The method of claim **17**, further comprising:

providing, by the one or more processors, the network allocation map to a third node, the third node being in a third domain, wherein the first node and the third node are connected and establish a fault tolerant configuration. 25

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